

Introduction to cognition:

Examples of cognitive processes (non-exhaustive)

- Decision making
 - Short-term memory / working memory
 - Learning
 - Long-term memory
 - Generalization
- Today
- Later
in
Unit

Cognition allows:

- The current brain to account for the past.
- The current brain to anticipate and reshape the future

Working definition: Cognition is what decouples the brain from the present.

Cognition is multifaceted and becomes more approachable when broken down.

Other topics not treated here (non-exhaustive)

- Perception
- Language
- Social behavior
- Attention

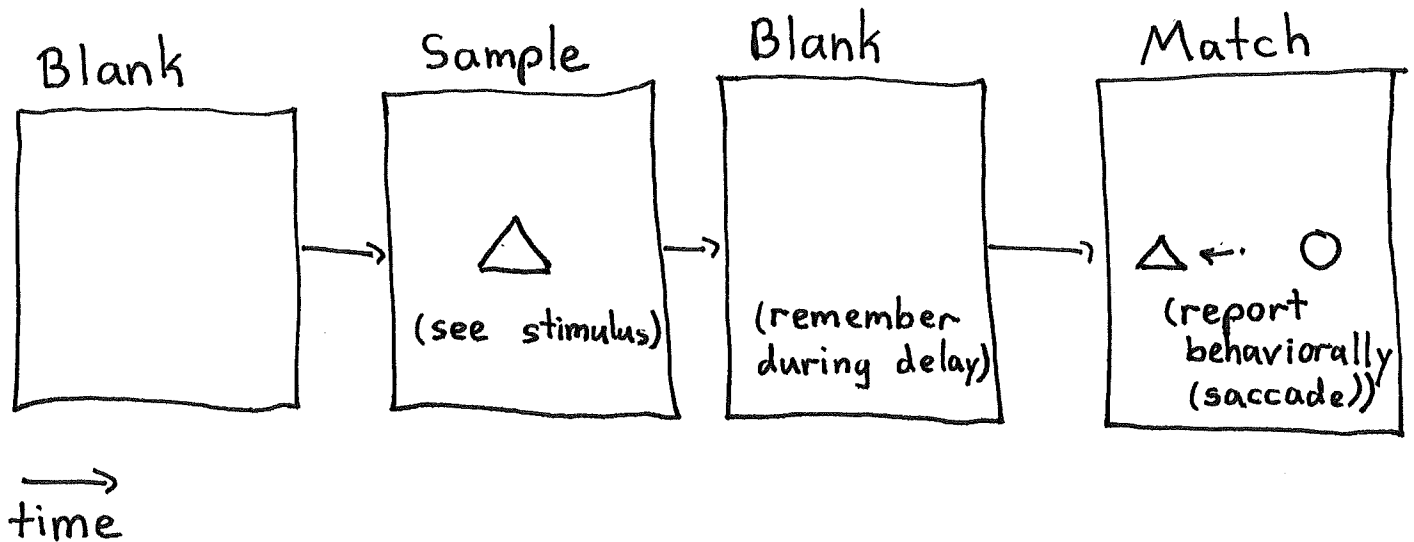
Short-term memory:

Memory: A trace of past events in the brain.

Short-term memories:

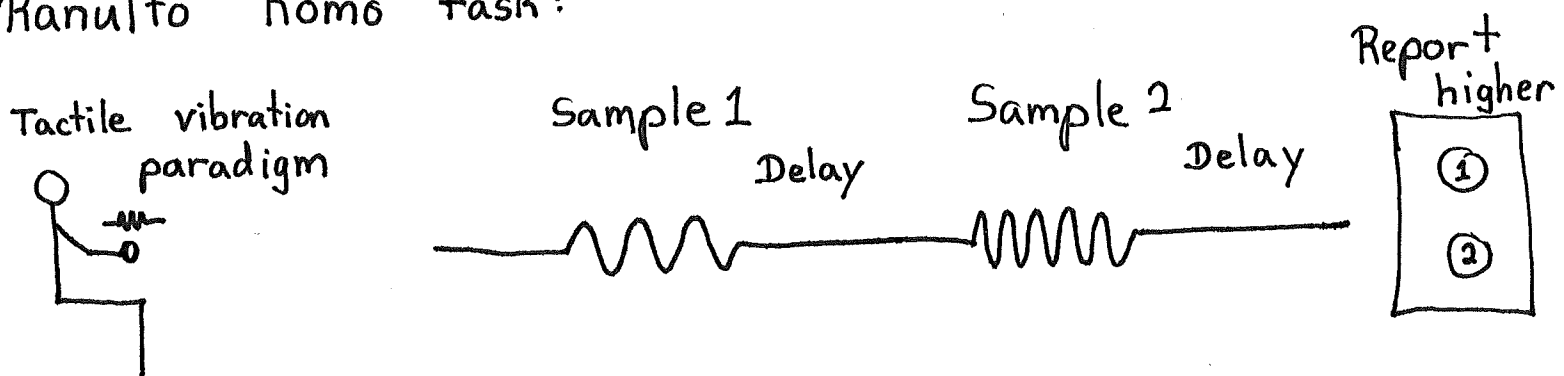
- Allows brain to account for what just happened.
- Does not require synaptic plasticity.
- Can be converted to long-term memory through synaptic plasticity.

Delayed match to sample task

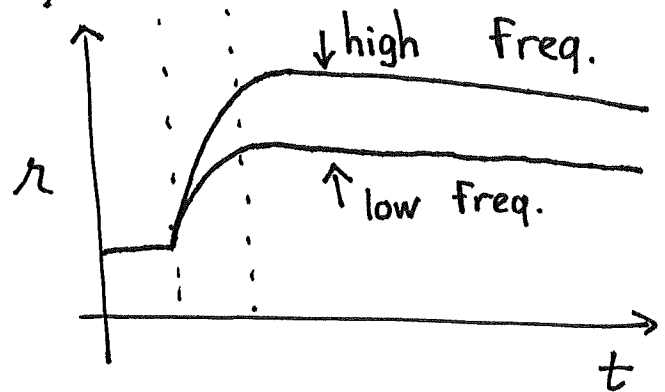
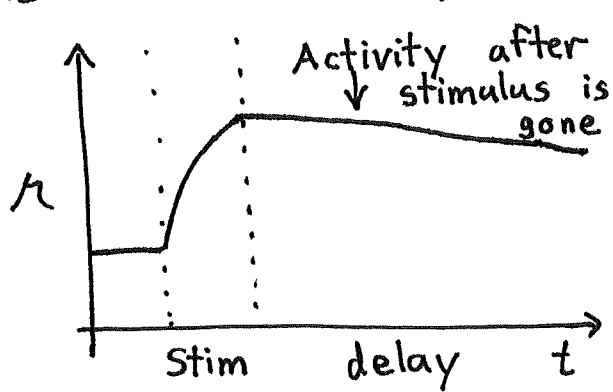


Parametric working memory: store a continuous value and later use it in a computation.

Ranulfo Romo task:

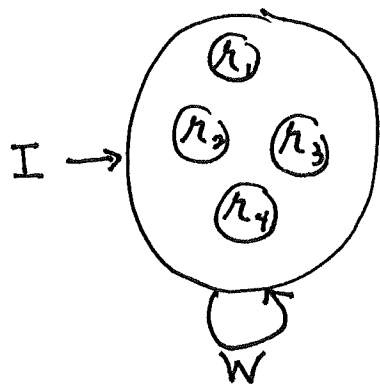


Persistent Neural Activity: Prefrontal cortex, etc.



Patterns discernible in single cells and more pronounced in population modes (e.g., PCA, etc.).

Persistent activity in linear neural networks



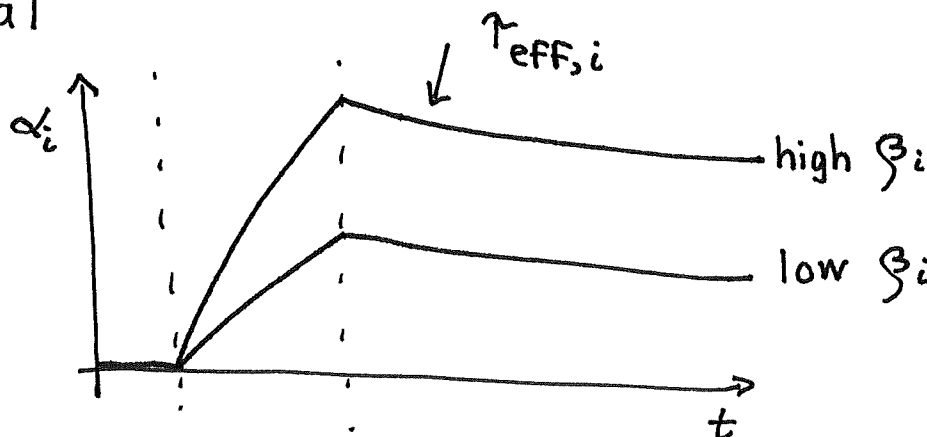
$$\tau \frac{dh_i}{dt} = -h_i + \sum_{j=1}^N W_{ij} h_j + I_i$$

In eigenbasis of W

$$\tau \frac{d\alpha_i}{dt} = -\alpha_i + \lambda_i \alpha_i + \beta_i$$

$$\underbrace{\tau}_{\tau_{\text{eff}, i}} \frac{d\alpha_i}{dt} = -\alpha_i + \underbrace{\frac{\beta_i}{1-\lambda_i}}_{\beta_{\text{eff}, i}}$$

If $\lambda_i \lesssim 1$, then mode response is a leaky integral

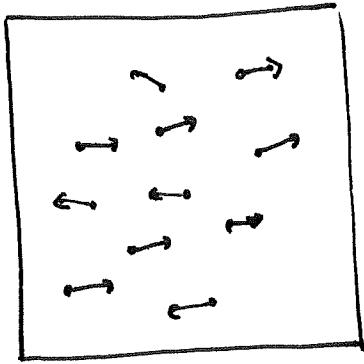


This requires no plasticity, just network dynamics.

Decision Making:

The Romo task is an example of decision making. A or B?

Also recall random dots task (Shadlen, Newsome)

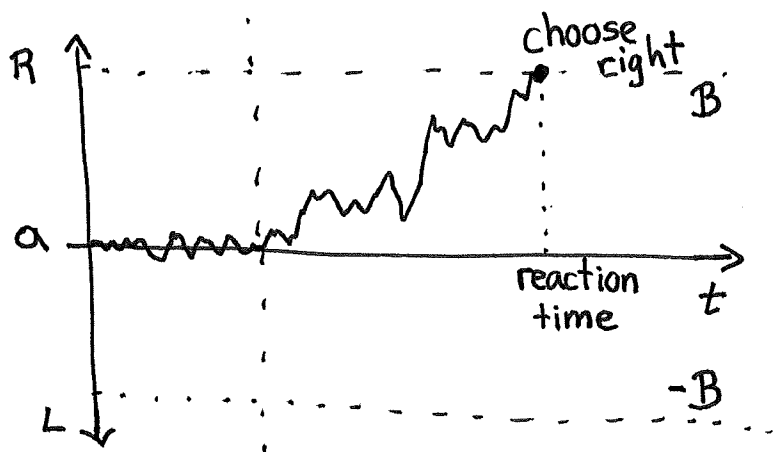


Decision: Left or Right

Hard if coherence is low

Evidence can be accumulated over time.

Drift - Diffusion Model of behavior



a: decision variable

B: decision bound

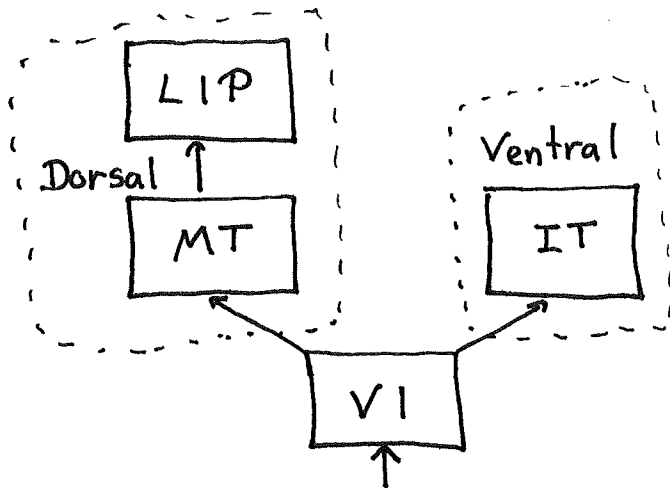
Basic equation:

$$\tau_a \frac{da}{dt} = \underbrace{-a}_{\text{Leaky integration}} + \underbrace{\gamma_R(t)}_{\text{R. evidence rate}} - \underbrace{\gamma_L(t)}_{\text{L. evidence rate}} + \underbrace{\gamma(t)}_{\text{random noise}}$$

$$a(0) = \underbrace{b}_{\text{Bias}} ; \quad a(\underbrace{T}_{\text{Decision time}}) = \pm B$$

Many related variants: collapsing bounds? Sticky bounds?
Go cue?

Utility of behavior model: reveals variables needing neural correlates.



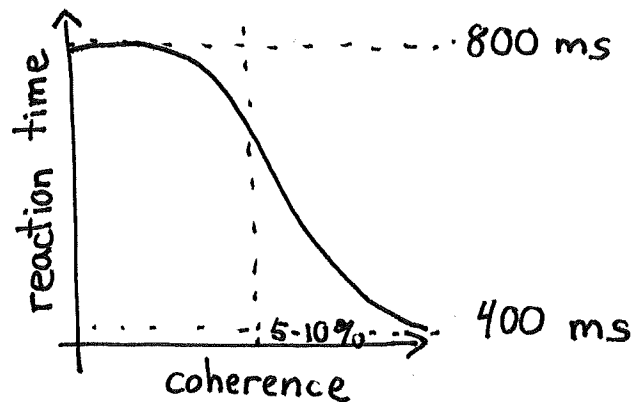
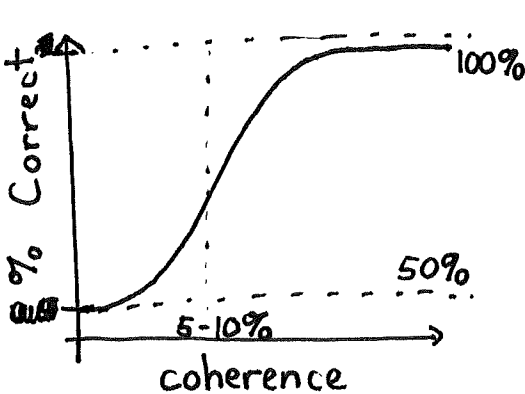
γ_R : R-selective MT neurons

γ_L : L-selective MT neurons

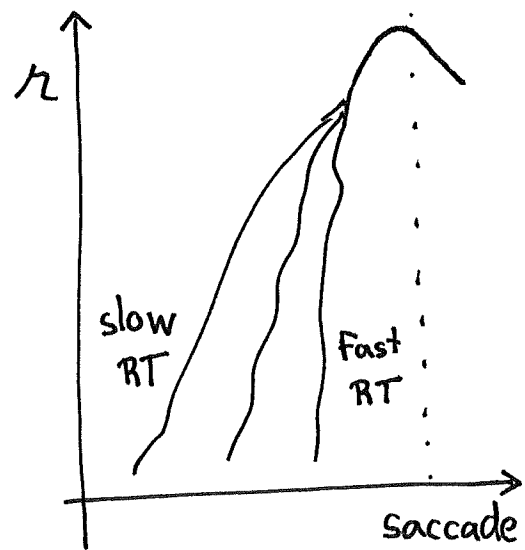
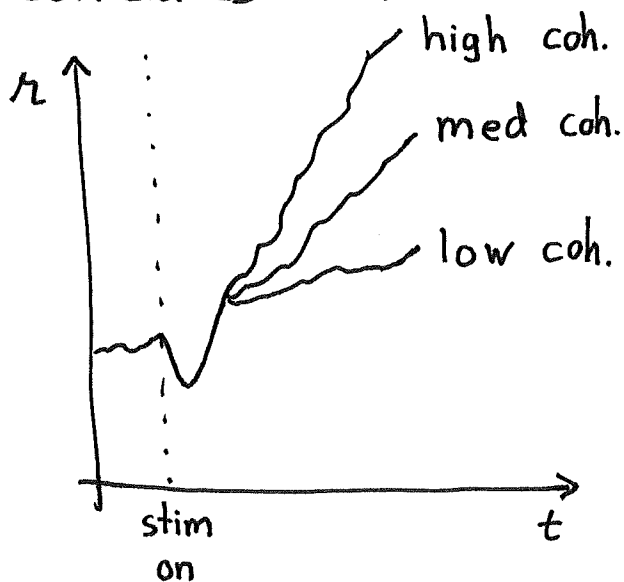
γ : stimulus variability & other neural variability

α : Neurons in LIP

Basic behavior quantification:



Neural correlates in LIP



Neural activity in LIP looks like " α ." More sophisticated behavior models $\Rightarrow$ more cognitive neural correlates.

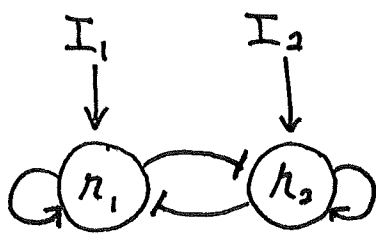
Just like short-term memory, evidence accumulation requires long integration time constants. Similar network mechanisms apply:

$$\tau_a \approx \frac{\tau}{1 - \lambda}$$

large eigenvalue

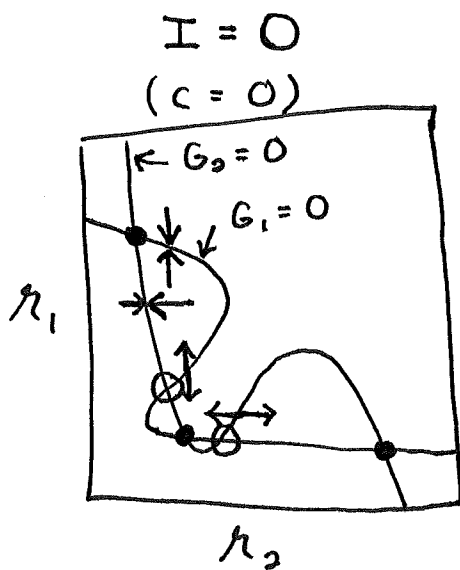
Nonlinear neural network models can also be devised for robust decision making.

Wong and Wang Model: Reduced biophysically realistic spiking neural network to 2D firing rate model.



$$\tau \frac{dr_i}{dt} = \underbrace{G_i(r_1, r_2, I)}_{\text{nonlinear functions}}$$

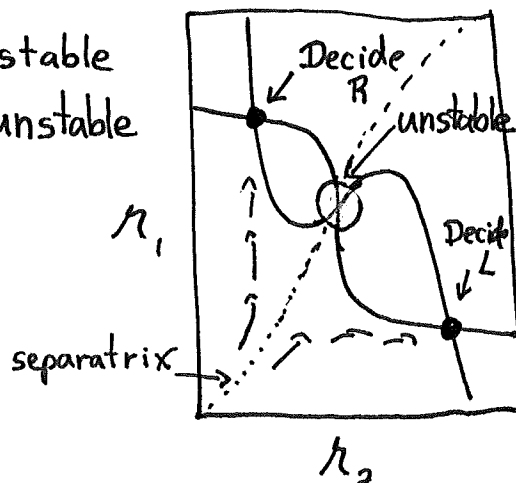
Phase-plane analysis: Two cubic-like nullclines



Both low;
One high

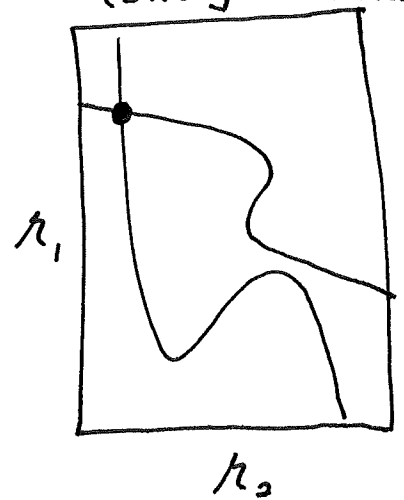
$I_1 > I_2$
(weak coherence)

• stable
○ unstable



Stochastic
Choice

$I_1 \gg I_2$
(strong coherence)



Choose "1"